(Please write your Enrollment Number) Enrollment No._____

End-Term Examination

(CBCS)(SUBJECTIVE TYPE)(OffLine)

Course Name:<B.Tech>, Semester:<VII>

(December, 2024)

Subject Code: BAI 403	**Subject: Computer Vision**
Time :3 Hours	**Maximum Marks :60**

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		**(2.5*8=20)**	**CO Mapping**
	a) Name the two key stages of computer vision and their use in understanding an image.		CO1
	b) Difference between Stochastic Gradient Descent & batch Gradient Descent.		CO1
	c) What is batch normalization in CNN?		CO2
	d) Why is pooling layer important? Explain with example.		CO3
	e) What is Attention Mechanism? What are it's types?		CO3
	f) Explain any 4 applications of Vision Transformers.		CO4
	g) Describe Semantic Segmentation. State it's applications.		CO4
	h) Explain Adversarial Network in details		**CO Mapping**
	UNIT I		
Q2	(a) What is "K" in the K-nearest neighbors algorithm? Why is the odd value of "K" preferred over even values in the KNN Algorithm? (b) How does Backpropagation Algorithm work? Describe the two major steps involved.	(10)	CO1
Q3	(a) What is optimization function of the loss function? Explain it's role in CV. (b) Explain Mean Squared Error and list it's applications.	(10)	CO1
	UNIT II		**CO Mapping**
Q4	(a) Explain AlexNet architecture. (b) Explain K-means clustering for vector quantization.	(10)	CO2
Q5	(a) Describe various weight initialization techniques in CNN. (b) Explain the role of regularization in CV. (c) Differentiate between Overfitting and Underfitting.	(10)	CO2
	UNIT III		**CO Mapping**
Q6	(a) Explain RNN architecture. (b) What are the different variants of RNN? Describe them. (c) Differentiate between feedforward Neural Networks and RNNs.	(10)	CO3
Q7	(a) How do Vision Language models work? (b) What is Image Captioning? Explain it's step by step workflow.	(10)	CO3
	UNIT IV		**CO Mapping**
Q8	(a) Explain Object Detection and it's working. (b) Explain different data Augmentation Techniques in details. (c) Differentiate between Two stage detectors and Single stage detectors.	(10)	CO4
Q9	(a) How does face recognition take place using Siamese Neural Networks? (b) What are conditional GANs? What are it's applications? (c) What are the use cases of Variational Autoencoders?	(10)	CO4